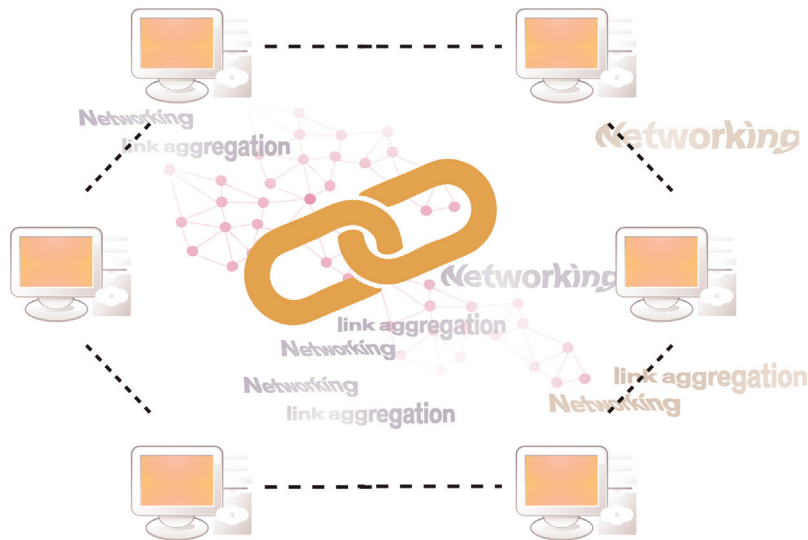


An Introduction to Link Aggregation in Networking

This article explains the concept of the link aggregation protocol in the world of networking and its benefits.



Link aggregation refers to the numerous methods of combining multiple network connections in parallel, in order to increase throughput beyond what a single connection could sustain, and to provide redundancy in case one of the links should fail. In order to understand the concept better, it is suggested that the reader be familiar with the basics of HTTP computer networking, which includes knowledge of the TCP/IP stack and of interconnecting devices like switches, routers or firewalls.

In Figure 1, two layer 3 switches are connected via a single 100Mbps link. If the link fails, communication between all hosts is lost. One solution to this problem is the creation of redundant links between switches. But we know that in case multiple links connect switches, a loop is formed.

So we need to enable the spanning tree protocol (STP) to overcome this issue. Spanning tree will define a link fault-tolerant design. But what if the bandwidth of 100Mbps is not able to handle the network load and we need to increase it? Replacing 100Mbps with 1Gbps may increase bandwidth. But all the devices will then need to be upgraded, which can be a costly affair.

The solution is to logically bundle multiple links to make them work as one single logical link, with an aggregated bandwidth of all links. For example, five such 100Mbps links can become a single 500Mbps link. This protocol is referred to as the link aggregation protocol. Each bundle of links is referred to as a link aggregated group (LAG). There can be one or more LAGs between switches or firewalls. This method is described by many

other terminologies like trunking, bonding or teaming. It encompasses all vendor-dependent proprietary solutions as well as standard solutions like LACP, etc. There can be many links between two devices, with or without LAG.

Link aggregation control protocol

The IEEE 802.3 working group has established a standard protocol for interoperable link bundling. Called the link aggregation control protocol or LACP (IEEE 802.3ad or 802.1ax), it provides dynamic configuration with auto negotiation of link bundling by sending LACP packets to peers. The maximum number of ports for bundling is eight (though it can vary in vendor-dependent implementations). 'Keep alive' packets are sent during runtime between link members.